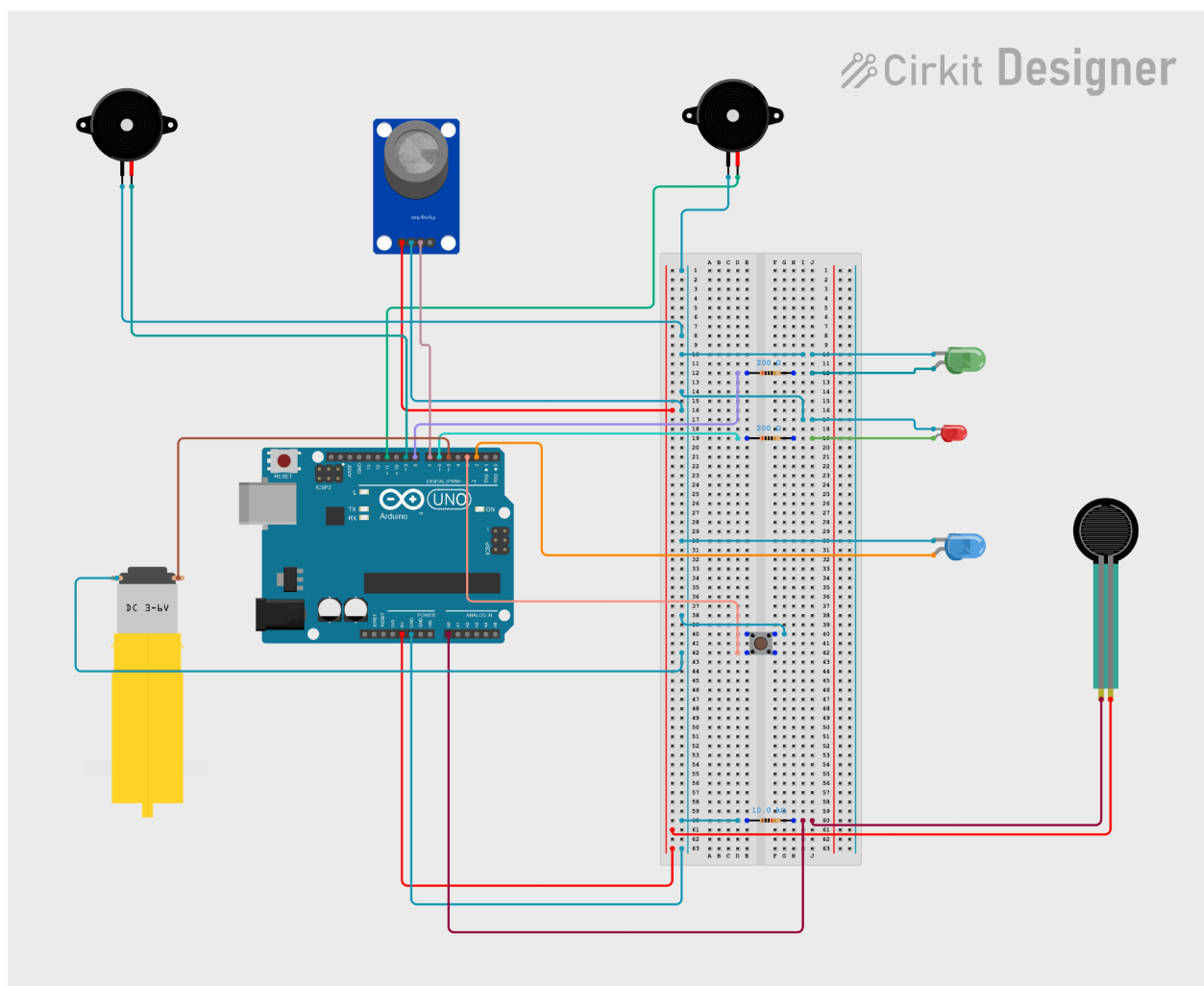


Circuit Documentation

Summary

This circuit is designed to perform two primary functions: smoke/gas detection and weight/theft monitoring. It utilizes an Arduino UNO microcontroller to control various components, including sensors, LEDs, a motor, and buzzers. The smoke/gas detection system activates an alarm and a motor when gas is detected, while the weight/theft monitoring system triggers an alarm if a significant weight drop is detected.



Component List

1. **Arduino UNO**
 - **Description:** A microcontroller board based on the ATmega328P.
 - **Pins:** UNUSED, IOREF, Reset, 3.3V, 5V, GND, Vin, A0, A1, A2, A3, A4, A5, SCL, SDA, AREF, D13, D12, D11, D10, D9, D8, D7, D6, D5, D4, D3, D2, D1, D0.
2. **DC Gear Motor (3-6V)**
 - **Description:** A motor used to simulate an exhaust fan.
 - **Pins:** V+, V-.
3. **Piezo Speaker**
 - **Description:** Used as an alarm buzzer.
 - **Pins:** pin1, pin2.
4. **LED Two Pin (Green)**
 - **Description:** Indicates no gas detected.
 - **Pins:** Cathode, Anode.
5. **LED Two Pin (Red)**
 - **Description:** Indicates gas detected.
 - **Pins:** Cathode, Anode.
6. **Resistor (200 Ohms)**
 - **Description:** Current limiting resistor for LEDs.
 - **Pins:** pin1, pin2.
7. **Force Sensor**
 - **Description:** Used to detect weight changes.
 - **Pins:** pin 1, pin 2.
8. **Resistor (10k Ohms)**
 - **Description:** Pull-down resistor for the force sensor.
 - **Pins:** pin1, pin2.
9. **Pushbutton**
 - **Description:** Used to arm/disarm the weight/theft system.
 - **Pins:** Pin 3 (out), Pin 4 (out), Pin 1 (in), Pin 2 (in).
10. **LED Two Pin (Blue)**
 - **Description:** Indicates armed status for the weight/theft system.
 - **Pins:** Cathode, Anode.
11. **MQ-2 Gas Sensor**
 - **Description:** Detects smoke/gas presence.
 - **Pins:** VCC, GND, DO, AO.

Wiring Details

Arduino UNO

- **5V:** Connected to Force Sensor pin 2 and MQ-2 Gas Sensor VCC.
- **GND:** Connected to:
 - Resistor (10k Ohms) pin1
 - Pushbutton Pin 1 (in)
 - LED Two Pin (Blue) Cathode
 - Piezo Speaker pin1

- MQ-2 Gas Sensor GND
- LED Two Pin (Green) Cathode
- LED Two Pin (Red) Cathode
- DC Gear Motor V-
- Piezo Speaker pin1
- **A0**: Connected to Force Sensor pin 1 via Resistor (10k Ohms) pin2.
- **D2**: Connected to LED Two Pin (Blue) Anode.
- **D3**: Connected to Pushbutton Pin 4 (out).
- **D5**: Connected to DC Gear Motor V+.
- **D6**: Connected to Resistor (200 Ohms) pin1, which is connected to LED Two Pin (Red) Anode.
- **D7**: Connected to MQ-2 Gas Sensor DO.
- **D8**: Connected to Resistor (200 Ohms) pin1, which is connected to LED Two Pin (Green) Anode.
- **D9**: Connected to Piezo Speaker pin2.
- **D11**: Connected to Piezo Speaker pin2.

DC Gear Motor (3-6V)

- **V+**: Connected to Arduino UNO D5.
- **V-**: Connected to Arduino UNO GND.

Piezo Speaker

- **pin1**: Connected to Arduino UNO GND.
- **pin2**: Connected to Arduino UNO D9 and D11.

LED Two Pin (Green)

- **Cathode**: Connected to Arduino UNO GND.
- **Anode**: Connected to Resistor (200 Ohms) pin2.

LED Two Pin (Red)

- **Cathode**: Connected to Arduino UNO GND.
- **Anode**: Connected to Resistor (200 Ohms) pin2.

Resistor (200 Ohms)

- **pin1**: Connected to Arduino UNO D6 and D8.
- **pin2**: Connected to LED Two Pin (Green) Anode and LED Two Pin (Red) Anode.

Force Sensor

- **pin 1**: Connected to Resistor (10k Ohms) pin2 and Arduino UNO A0.
- **pin 2**: Connected to Arduino UNO 5V.

Resistor (10k Ohms)

- **pin1:** Connected to Arduino UNO GND.
- **pin2:** Connected to Force Sensor pin 1 and Arduino UNO A0.

Pushbutton

- **Pin 1 (in):** Connected to Arduino UNO GND.
- **Pin 4 (out):** Connected to Arduino UNO D3.

LED Two Pin (Blue)

- **Cathode:** Connected to Arduino UNO GND.
- **Anode:** Connected to Arduino UNO D2.

MQ-2 Gas Sensor

- **VCC:** Connected to Arduino UNO 5V.
- **GND:** Connected to Arduino UNO GND.
- **DO:** Connected to Arduino UNO D7.

Documented Code

The code for this circuit is divided into two main sections: smoke/gas detection and weight/theft monitoring. Below is a summary of the code functionality:

Smoke/Gas Detection

- **Pins Used:**
 - smokePin (D7): Reads the digital output from the MQ-2 Gas Sensor.
 - redLedPin (D6): Controls the Red LED to indicate gas detection.
 - greenLedPin (D8): Controls the Green LED to indicate no gas detection.
 - motorPin (D5): Activates the motor (exhaust fan) when gas is detected.
 - buzzer1Pin (D9): Activates the buzzer for smoke alarm.
- **Logic:**
 - The system continuously checks the smoke sensor. If gas is detected, the red LED and motor are activated, and the buzzer sounds. If no gas is detected, the green LED is activated, and the motor and buzzer are turned off.

Weight/Theft Monitoring

- **Pins Used:**
 - sensorPin (A0): Reads the analog value from the Force Sensor.
 - buzzer2Pin (D11): Activates the buzzer for weight alarm.
 - weightLed (D2): Indicates the armed status of the system.
 - btnPin (D3): Used to arm/disarm the system via a pushbutton.

- **Logic:**
 - The system monitors the weight using the force sensor. If the weight drops below a certain threshold, the alarm is triggered. The system can be armed or disarmed using the pushbutton.
- **Button Functions:**
 - `singleclick_Func()`: Arms the system if the weight is above the minimum threshold.
 - `doubleclick_Func()`: Disarms the system.

The code is structured to run both systems concurrently, using non-blocking timers and state machines to manage the logic efficiently.